Re-implementation of Visual SLAM in Dynamic Environments Based on Object Detection and Scene Flow

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1 Introduction

Simultaneous Localization and Mapping (SLAM) is a core capability for autonomous navigation. In particular, visual SLAM using lightweight cameras has gained traction. However, most SLAM systems assume a predominantly static scene and can fail in dynamic environments with moving objects. Simply treating dynamic features as outliers and discarding them wastes valuable information. Reliably identifying dynamic elements is itself challenging, as some detected objects may actually be static. Utilizing dynamic objects for joint camera and object pose estimation has the potential to improve accuracy and robustness. However, this requires careful association of object detections across frames and principled motion modeling. Our work re-implements the system proposed by Li et al.[4] to achieve accurate and stable visual SLAM in dynamic environments. The key insights are: Combining learning-based object detection with geometric scene flow estimation, Probabilistically discriminating between truly dynamic and static-but-potentially-dynamic objects, Tightly integrating dynamic object tracking in the SLAM optimization backend.

2 Related Work

For multi-object visual SLAM, the annotation of dynamic objects or features by detecting the continuous image stream is the basis for handling dynamic environments.

One approach is to combine object detection with multi-view geometry which uses semantic labeling to identify dynamic potential dynamic objects and using optical or scene flow to determine whether the object is in motion. For example, Zhang et al. [11] proposed a Visual Dynamic Object-aware (VDO) SLAM system, which combines instance segmentation with optical flow. It proposed to use dense optical flow in the areas marked as dynamic to achieve more stable tracking of dynamic objects, resulting in a large overall computation.

The first class of methods decouples dynamic object tracking from SLAM as two independent estimators. Barsan et al. [1] proposes a dense mapping algorithm based on stereo vision for urban environments, which can reconstruct both static backgrounds and moving objects. However, the main drawback of these methods is that their accuracy is highly dependent on the precision of camera pose estimation.

3 Data Description

The KITTI dataset is a pioneering resource in the field of autonomous driving. The dataset features real-world driving scenarios captured in urban and rural environments, providing valuable data for tasks such as object detection, tracking, and depth estimation. Its comprehensive annotations and diverse scenarios make it a fundamental benchmark in evaluating the performance of autonomous driving systems. We extract sequences from the Visual Odometry dataset. For each sequence, we retain the first 200 images for evaluation.

4 Baseline Method

The traditional Visual SLAM includes five steps:
1) Initialization: This step involves setting up a coordinate system or reference frame in the environment. It establishes the initial conditions for the SLAM system to start mapping and localizing.
2) Tracking: During tracking, the SLAM system continuously estimates the camera's position and orientation as it moves through the environment.
3) Mapping: The SLAM system constructs or updates a representation of the environment based on the tracked camera poses and the visual features observed in the images.
4) Re-localization: Re-localization is the process of initialising the SLAM system's tracking and mapping capabilities if the camera loses track of its position or if tracking fails due to occlusions or sudden movements.
5) Global Map Optimization: Global map optimization aims to refine the map and camera poses to compensate for accumulative error in the estimated camera poses and map representation due to noise, drift, or inaccuracies.
One primary method is the feature-based method. In feature-based visual SLAM, the system extracts distinctive features from the images captured by the camera, such as corners, edges, or keypoints. These features are then matched across different frames to estimate the motion of the camera and build a map of the environment.

5 Propose Method

We started by processing a set of stereo images from the KITTI dataset. For "Object Detection and Feature Discrimination," we employed YOLO to detect objects on stereo images and categorize feature points as either Static or Potential Dynamic Points. "Feature Extraction" involves detecting features using the ORB algorithm on the stereo images, followed by feature matching with the Flann matcher, a bidirectional check and RANSAC.

During "Initial Pose Estimation," we calculated and minimized the reprojection error using least squares optimization. For the "Kullback-Leibler Divergence" step, we used KL divergence to assess the difference between predicted and observed point descriptors, facilitating the distinction between Dynamic and Potential Dynamic Objects. After isolating feature points on non-moving objects, we incorporated these points to refine the optimization of the transformation matrices

5.1 Object Detection and Feature Discrimination

Object detection is a crucial step in the Visual SLAM, as it helps identify and localize objects of interest in the captured images. In our project, the YOLOv8n[3] object detection model is employed to detect objects in the left and right images of a stereo camera setup. The YOLOv8n model is pretrained on a large dataset and can detect a wide range of object categories. By applying the pretrained model to the input images, it can identify and localize objects present in the scene. The model takes the left and right images as input and outputs the bounding box coordinates of the detected objects in each image. These bounding boxes provide valuable information about the location and extent of the objects in the scene.

Two approaches are tested for feature discrimination in our project. One is with K-means[9] and the other is without K-means clustering. When using K-means clustering, the depths of the feature points associated with the detected objects are clustered to separate static and dynamic points based on their similarity in depth. This is based on the assump-

tion from the paper that dynamic objects always appear in front of the static background, meaning that the depth of features on dynamic objects is always much smaller than that of static background features. On the other hand, feature discrimination without K-means clustering relies solely on the spatial relationship between the feature points and the detected object bounding boxes. Feature points inside the bounding boxes are considered dynamic, while those outside are treated as static. After testing both the method, we used the second method, which does not involve K-means clustering, for feature discrimination since it gave better results.



Figure 1: Filtering static and potential dynamic features

5.2 Feature Extraction

Feature extraction serves as a foundational element in the Visual SLAM and the first step is detecting features. The feature detection method we select is the ORB detection algorithm [8], a more computationally efficient alternative to SIFT.

Following feature detection, the next step involves matching features between left and right images for 3D reconstruction, as well as across time frames to establish camera pose transformations. We employ the FLANN matching algorithm [7], which efficiently identifies approximate nearest neighbors among the detected features, greatly accelerating the matching process. To verify the quality and accuracy of these matches, we apply Lowe's ratio test [5], filtering out less reliable matches by comparing the distance between the nearest and second-nearest neighbors. We further refine these matches using the RANSAC algorithm [2], based on the fundamental matrix, to robustly estimate geometric consistency. This same combination of FLANN, the ratio test, and RANSAC ensures fast and robust feature matching for both

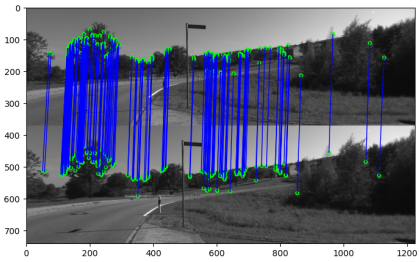


Figure 2: Feature matching between two consecutive time frames

left-right image matching and temporal frame alignment.

5.3 Initial Pose Estimation

Before applying KL Divergence to distinguish between dynamic and potential dynamic features, we first need to calculate an initial transformation matrix using only static features. We obtain this matrix by minimizing the reprojection errors between two consecutive time frames, i and j, followed by applying RANSAC to find the optimized transformation matrix.

For each matching pair of feature points $[p_k^{I_j}, P_k^{c_i}]$ derived from the previous calculations, where $p_k^{I_j}$ is the kth 2D point in the image frame of frame j and $P_k^{c_i}$ is the kth 3D point in the camera frame of frame i, and given the known camera pose $T_{c_i}^w$ for frame i, the reprojection error can be expressed as:

$$r_k = p_k^{I_j} - \Pi((T_{c_j}^w)^{-1} T_{c_i}^w P_k^{c_i}) \tag{1}$$

where Π represents the reprojection function for a rectified stereo camera that projects 3D points in the camera frame to 2D points in the image frame. By minimizing these reprojection errors across all pairs of feature points, we determine the initial pose $T_{c_j}^w$.

For this purpose, we utilized the Levenberg-Marquardt algorithm [6], an optimization method that synergistically combines gradient descent and Gauss-Newton techniques. This algorithm adjusts a damping factor to iteratively minimize the sum of

squared function differences between observed data and a model, effectively refining our estimations.

To enhance the accuracy of our initial pose estimation in the presence of potential outliers, we incorporated the RANSAC algorithm into our approach. RANSAC repetitively selected random subsets of the matched features, used the Levenberg-Marquardt algorithm to calculate the transformation matrix, and retained the solution with the least error. This integration is crucial as it significantly improves the reliability and precision of our pose estimations.

5.4 Kullback-Leibler Divergence[10]

Using YOLO for object detection, we differentiated between static feature points (outside object bounding boxes) and potential dynamic feature points (inside bounding boxes). These dynamic points might be on stationary or moving objects. In visual SLAM, static points are used to compute transformation matrices.

We used Kullback-Leibler (KL) divergence to assess the difference between two probability distributions derived from feature point descriptors. A high KL divergence indicates a significant difference, suggesting movement, while a low value implies similarity, indicating no movement. Specifically, we compared predicted feature points, transformed from previous to current frames, with actual observed points. By normalizing these descriptors into probability distributions and calculating their KL divergence, we found that dynamic objects tend to show higher entropy in these comparisons than static ones.

5.5 Local Pose Estimation

After calculating entropies between predicted potential dynamic feature points and observed potential dynamic feature points using KL divergence, we calculate the means of KL divergence values of each bounding box containing each object. Suppose the mean of the KL divergence value of feature points is higher than 0.3. In that case, we regard the object inside of the bounding box as a dynamic object. If the bounding box contains the feature points whose mean is lower than 0.3, we added those feature points to the static feature points we found in the YOLO part. After that, we calculate the transformation matrix again to compare the performance with other cases: calculate trajectory using all feature points and using only static feature points (outside of bounding boxes)

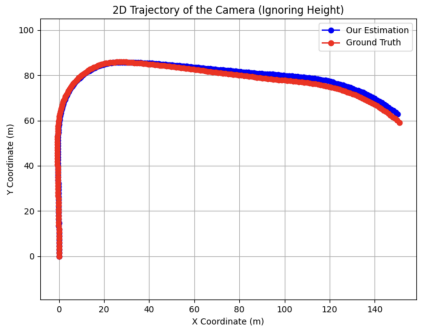


Figure 3: Camera trajectory of the sequence 02 with only static features

6 Result-Evaluation

The evaluation metrics employed are Absolute Trajectory Error (ATE) and Relative Pose Error (RPE). ATE, defined as the root mean square error of error matrices, quantifies the average deviation from the ground truth trajectory per frame. RPE evaluates the local trajectory accuracy over a set time interval, reflecting the trajectory's drift.

Fig.3 and 4 illustrate the outcomes with only static features, while Fig.5 and 6 display the results after applying KL divergence.

7 Challenges

YOLO is a real-time object detection algorithm that can detect and localize objects in images efficiently. However, it may struggle with varying object scales and orientations. Additionally, YOLO may generate false positives by mistakenly identifying non-object regions as objects, or false negatives by failing to detect objects that are present. These errors can propagate to the subsequent feature discrimination stage affecting the accuracy of static and dynamic feature

KITTI Odometry	Every Feature Points			
	ATE Translation (m)	ATE Rotation (rad)	RPE Translational (m)	RPE Rotational (rad)
0	67.0798	0.7067	0.4666	0.0114
2	67.2220	1.1715	0.5664	0.0119
5	75.4940	0.5273	0.5661	0.0091
8	13.5966	1.0875	0.1278	0.0096
10	18.1455	1.6946	0.2924	0.0140

KITTI Odometry	Only Static Feature Points			
	ATE Translation (m)	ATE Rotation (rad)	RPE Translational (m)	RPE Rotational (rad)
0	1.6112	0.0150	0.0309	0.0009
2	1.1368	0.0150	0.0404	0.0005
5	1.4383	0.0058	0.0403	0.0004
8	20.8001	1.5000	0.5391	0.0130
10	34.3555	2.1749	0.5170	0.0167

KITTI Odometry	Static Feature Points + Feature Points on Static Objects			
	ATE Translation (m)	ATE Rotation (rad)	RPE Translational (m)	RPE Rotational (rad)
0	1.7369	0.0139	0.0386	0.0009
2	1.3787	0.0194	0.0391	0.0005
5	1.5500	0.0047	0.0309	0.0004
8	28.4557	0.7335	1.1795	0.0068
10	42.9194	2.2822	0.5018	0.0165

Table 1: Absolute Trajectory Error (ATE) and Relative Pose Error (RPE)

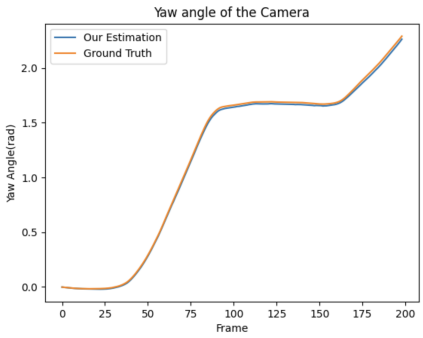


Figure 4: Camera yaw angle of the sequence 02 with only static features

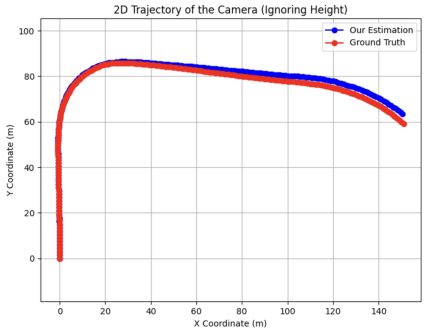


Figure 5: Camera yaw angle of the sequence 02 after applying KL divergence

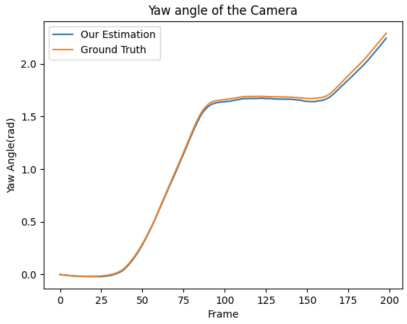


Figure 6: Camera yaw angle of the sequence 02 after applying KL divergence

point classification.

When using K-means clustering for feature discrimination, the number of clusters is typically set to two, representing static and dynamic points. The main challenge lies in accurately separating feature points into these two clusters based on their depth similarity. In complex scenes with multiple moving objects at various depths, the depth distributions of static and dynamic points may overlap, making it difficult to achieve a clear separation. On the other hand, when performing feature discrimination without K-means clustering, the challenges involve accurately associating feature points with detected object bounding boxes. As we mentioned above, YOLO may generate false positives or false negatives. This problem causes consecutive problems using KL divergence output to discriminate feature points on dynamic objects. Failing objects correctly means the probability of detecting dynamic objects is higher. Moreover, the best threshold of the mean value of KL divergence values in each bounding box could vary image by image and sequence by sequence. Therefore, it is good to set the threshold strictly, but then we cannot get a meaningful amount of new feature points.

8 Conclusion

In conclusion, the research conducted in this study reimplements visual SLAM systems for dynamic environments by integrating YOLO-based object detection with scene flow techniques. By distinguishing between static and potentially dynamic feature points and applying Kullback-Leibler divergence to assess the discrepancy between predicted and observed feature point descriptors, we expected that this approach would improve the accuracy of feature tracking in scenes with moving objects and enhance the robustness of SLAM applications. Still, we could not see much improvement because of the many challenges we encountered in object detection, K-means, and setting the universal threshold for KL divergence. However, suppose we can overcome the mentioned challenge and employ static and dynamic object tracking within the SLAM optimization framework. In that case, we can further contribute to a more accurate and reliable localization and mapping, especially in challenging dynamic settings. This future work could be a significant step forward in applying advanced visual SLAM techniques to real-world dynamic environments, which is crucial for developing autonomous systems capable of operating in complex and unpredictable conditions.

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